

Polluted membranes under high pressure: A neutron scattering study

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ABSTRACT

Plastic pollution is widely recognized for its visible impact on oceans and wildlife, but less attention is given to the molecular byproducts that arise as plastics degrade [1]. Among these, styrene oligomers (SOs)—small molecules released from polystyrene—may pose hidden risks to living systems, including under the high pressures of deep ocean environments where plastic degradation products have been detected [2]. We investigated how SOs interact with model cell membranes composed of dipalmitoylphosphatidylcholine (DPPC). Using neutron reflectivity (see fig.), neutron diffraction, and molecular dynamics simulations at both ambient and elevated pressures, we found that SOs disrupt the structural organization of lipid bilayers and weaken the membrane's fluid-to-gel phase transition. These results highlight how plastic-derived molecules can subtly compromise biomembrane stability and function, even under extreme conditions

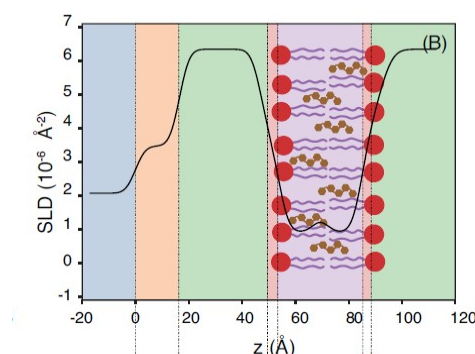


Fig. Scattering length density (SLD) profile from reflectivity data of a DPPC bilayer with 30% styrene oligomer.

REFERENCES

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